

# **Aprendizagem Aplicada à Segurança**

## SPAM Dectetion

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- The term “spam” is internet slang that refers to unsolicited commercial email (UCE).
- The first reported case of spam occurred in 1898, when the New York Times reported unsolicited messages circulating in association with an old swindle.
- The term “spam” was coined in 1994, based on a now-legendary Monty Python’s Flying Circus sketch, where a crowd of Vikings sings progressively louder choruses of “SPAM! SPAM! SPAM!”



# SPAM



Dear Sir,

I am prince [REDACTED] from Nigeria. Your help would be very appreciated. I want to transfer all of my fortune outside of Nigeria due to a frozen account, If you could be so kind and transfer small sum of 3 500 USD to my account, I would be able to unfreeze my account and transfer my money outside of Nigeria. To repay your kindness, I will send 1 000 000 USD to your account.

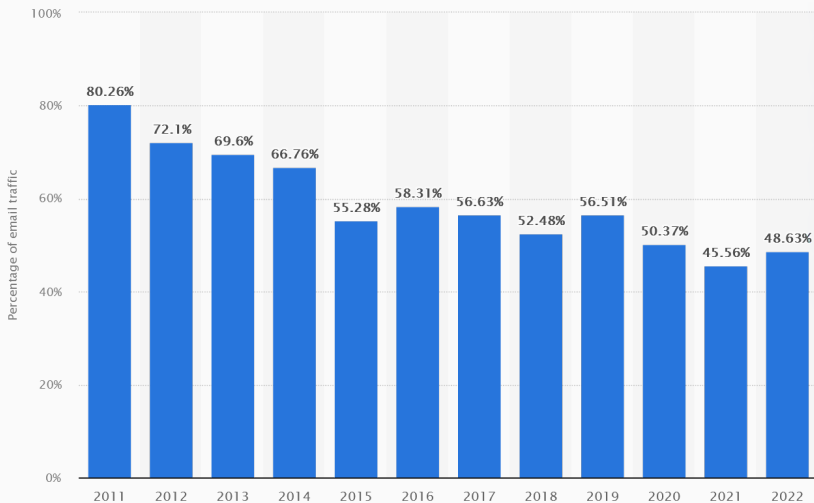
Please contact me to proceed

Prince [REDACTED]

# Fight against SPAM

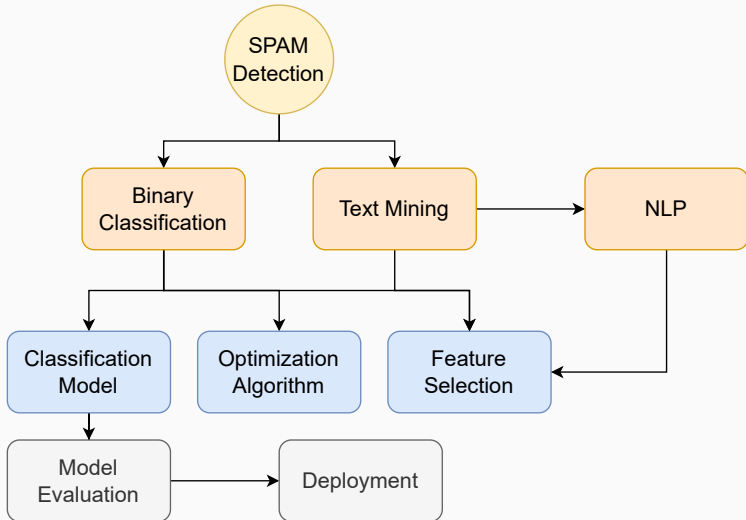
- *Huge* list of [anti-spam techniques](#)
- From common sense to *Bayesian spam filtering*
- Unfortunately it is a costly battle

# Fight against SPAM



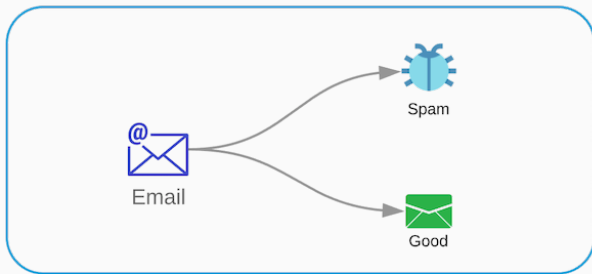


# SPAM Detection



# Binary Classification

- Binary classification is the task of classifying the elements of a set into two groups (each called class) on the basis of a classification rule.
- For this application one message can either be spam or ham.



Binary classification

- Text mining is the process of deriving high-quality information from text.
- Combines concepts from Machine Learning, Linguistic and statistical analysis.
- In this area we will explore the methods used to rank words/tokens and the BoW model.

## Bag of Words (Bow) model

|                    | the | red | dog | cat | eats | food |
|--------------------|-----|-----|-----|-----|------|------|
| 1. the red dog →   | 1   | 1   | 1   | 0   | 0    | 0    |
| 2. cat eats dog →  | 0   | 0   | 1   | 1   | 1    | 0    |
| 3. dog eats food → | 0   | 0   | 1   | 0   | 1    | 1    |
| 4. red cat eats →  | 0   | 1   | 0   | 1   | 1    | 0    |

# Natural Language Processing (NLP)

- NLP gives the computers the ability to understand text.
- Combines *Syntax* and *Semantic* into the analysis.
- One famous examples are the Large Language Models (LLMs) that power OpenAI Chat GPT.

# Classification Model

- SPAM detection is “considered” a toy example.
- As such, we will explore two of the simplest learning models: Naive Bayes and Logistic Regression.

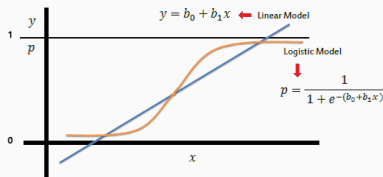
Likelihood of the Evidence given that the Hypothesis is True

Prior Probability of the Hypothesis

$$P(H|E) = \frac{P(E|H) * P(H)}{P(E)}$$

Posterior Probability of the Hypothesis given that the Evidence is True

Prior Probability that the evidence is True



# Model Evaluation

- Classification model can be evaluated using a confusing matrix
- The simplest methods to evaluate a model is through accuracy:  $acc = \frac{TP+TN}{TP+TN+FP+FN}$

|                 | Predicted Positive                  | Predicted Negative                                  |                                                   |
|-----------------|-------------------------------------|-----------------------------------------------------|---------------------------------------------------|
| Actual Positive | TP<br><i>True Positive</i>          | FN<br><i>False Negative</i>                         | Sensitivity<br>$\frac{TP}{(TP + FN)}$             |
| Actual Negative | FP<br><i>False Positive</i>         | TN<br><i>True Negative</i>                          | Specificity<br>$\frac{TN}{(TN + FP)}$             |
|                 | Precision<br>$\frac{TP}{(TP + FP)}$ | Negative Predictive Value<br>$\frac{TN}{(TN + FN)}$ | Accuracy<br>$\frac{TP + TN}{(TP + TN + FP + FN)}$ |

# Model Evaluation

